Kartikey Attri
8025320046
ME-CSE
Group-2

PCS223 Advanced Computer Networks

1. **To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.**

Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
  - Wi-Fi
  - Ethernet

Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately

Step 3: Generate Network Traffic
Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com

Step 4: Stop Capture

- Click the **red Stop button**
- Captured packets will remain on the screen

Step 5: Apply Basic Filters
Use the **Display Filter bar**:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets

Filter	Purpose
dns	Show DNS packets
icmp	Show ping packets

Step 6: Analyze a Packet

- Click on any packet
- Observe:
  - **Ethernet header** Ethernet II, Src: HewlettP_22:9d:ca (f4:39:09:22:9d:ca), Dst: BrocadeC_90:80:78 (74:8e:f8:90:80:78)
  - **IP header** Internet Protocol Version 4, Src: 172.16.93.50, Dst: 23.34.253.33
  - **TCP/UDP header** Transmission Control Protocol, Src Port: 64114, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
  - Payload data

Observation

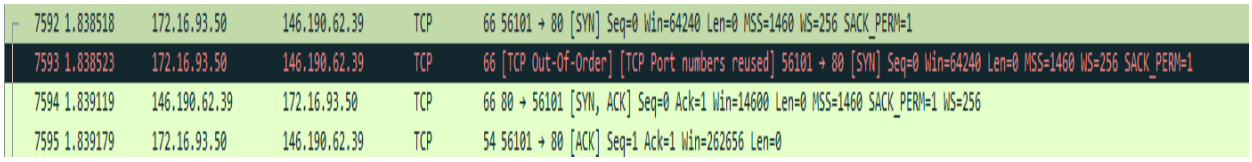
Interface used:

- Protocols observed: tcp
- Source and destination IP addresses

  Source Address: 172.16.93.50, Destination Address: 23.34.253.33
- Packet length : 54 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.



7592	1.838518	172.16.93.50	146.190.62.39	TCP	66 56101 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7593	1.838523	172.16.93.50	146.190.62.39	TCP	66 [TCP Out-Of-Order] [TCP Port numbers reused] 56101 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7594	1.839119	146.190.62.39	172.16.93.50	TCP	66 80 → 56101 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM=1 WS=256
7595	1.839179	172.16.93.50	146.190.62.39	TCP	54 56101 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0

```
C:\Users\CSED>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::73bb:fff6:dbe9:6e3d%10
   IPv4 Address. . . . . . . . . . . : 172.16.93.50
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Default Gateway . . . . . . . . . : 172.16.64.1

C:\Users\CSED>PING 172.16.64.1

Pinging 172.16.64.1 with 32 bytes of data:
Reply from 172.16.64.1: bytes=32 time=4ms TTL=128
Reply from 172.16.64.1: bytes=32 time=2ms TTL=128
Reply from 172.16.64.1: bytes=32 time=1ms TTL=128
Reply from 172.16.64.1: bytes=32 time=1ms TTL=128

Ping statistics for 172.16.64.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms
```